

Equation Sample 3

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

$$\begin{aligned} & E_{p(\tilde{x})}[\langle GNN_{\theta}(\tilde{x}), \nabla_{\tilde{x}} \log p(\tilde{x}) \rangle] \\ &= \int_{\tilde{x}} p(\tilde{x}) \langle GNN_{\theta}(\tilde{x}), \nabla_{\tilde{x}} \log p(\tilde{x}) \rangle d\tilde{x} \\ &= \int_{\tilde{x}} p(\tilde{x}) \left\langle GNN_{\theta}(\tilde{x}), \frac{\nabla_{\tilde{x}} p(\tilde{x})}{p(\tilde{x})} \right\rangle d\tilde{x} \\ &= \int_{\tilde{x}} \langle GNN_{\theta}(\tilde{x}), \nabla_{\tilde{x}} p(\tilde{x}) \rangle d\tilde{x} \\ &= \int_{\tilde{x}} \left\langle GNN_{\theta}(\tilde{x}), \nabla_{\tilde{x}} \left(\int_x p(\tilde{x}|x) p(x) dx \right) \right\rangle d\tilde{x} \\ &= \int_{\tilde{x}} \left\langle GNN_{\theta}(\tilde{x}), \int_x p(x) \nabla_{\tilde{x}} p(\tilde{x}|x) dx \right\rangle d\tilde{x} \\ &= \int_{\tilde{x}} \left\langle GNN_{\theta}(\tilde{x}), \int_x p(\tilde{x}|x) p(x) \nabla_{\tilde{x}} \log p(\tilde{x}|x) dx \right\rangle d\tilde{x} \\ &= \int_{\tilde{x}} \int_x p(\tilde{x}|x) p(x) \langle GNN_{\theta}(\tilde{x}), \nabla_{\tilde{x}} \log p(\tilde{x}|x) \rangle dx d\tilde{x} \\ &= E_{p(\tilde{x}, x)}[\langle GNN_{\theta}(\tilde{x}), \nabla_{\tilde{x}} \log p(\tilde{x}|x) \rangle] \end{aligned}$$